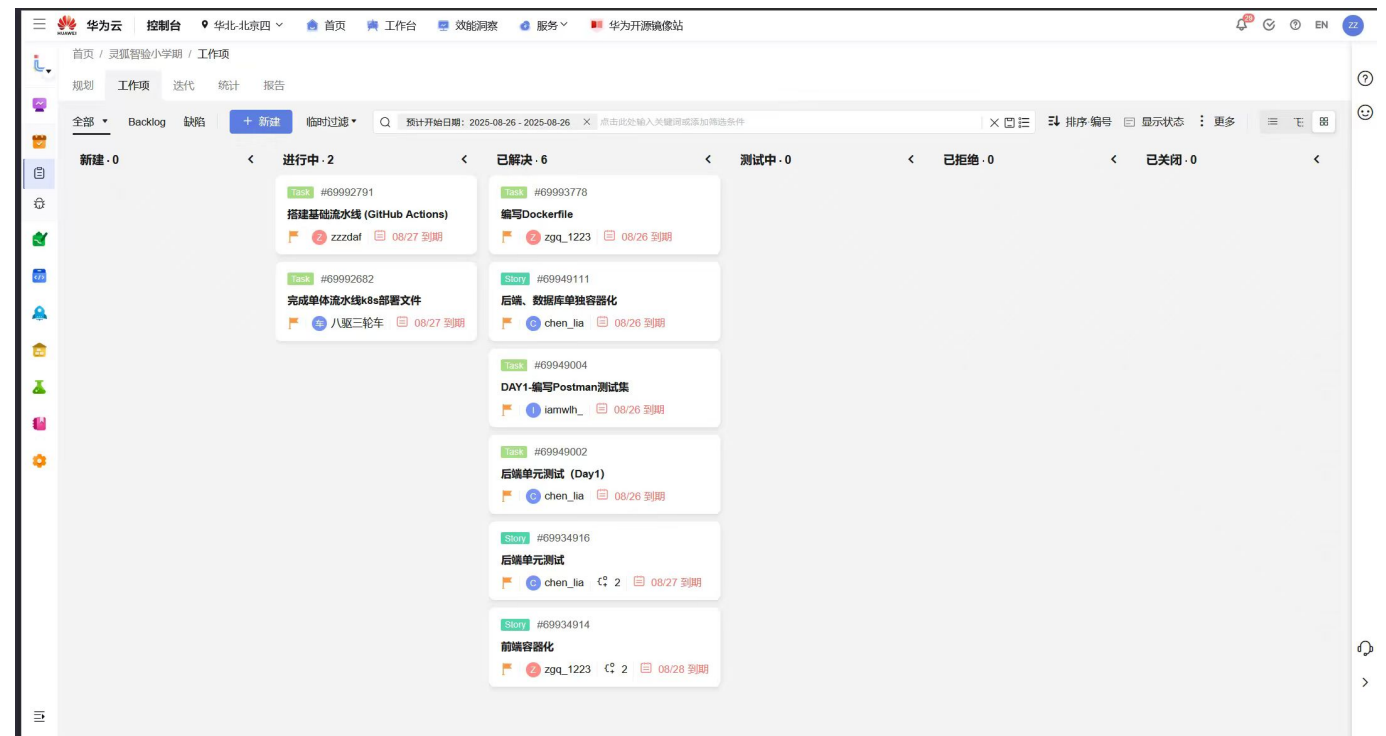


灵狐智验平台（12组）技术总结报告

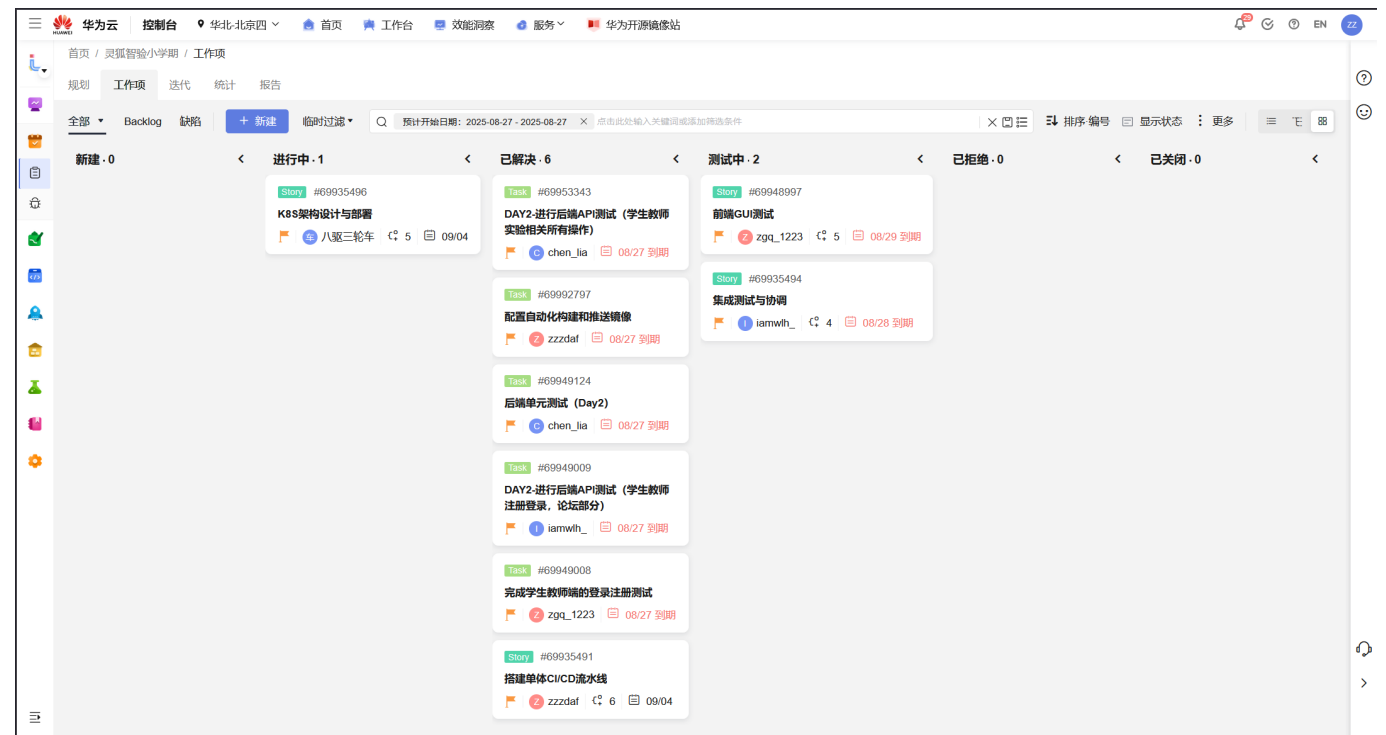
成员：王露函、张宸菲、陈利雅、张管倩、马浩龄

项目管理截屏

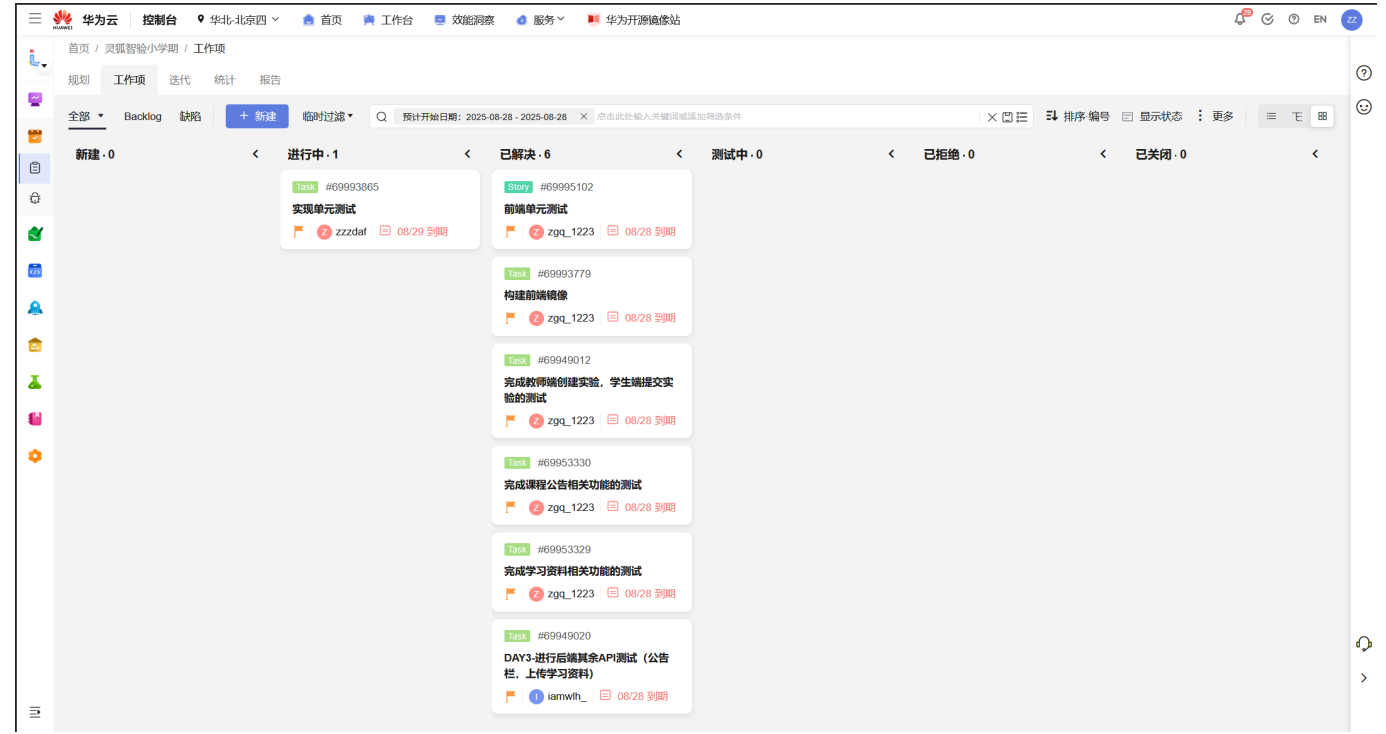
8.26:



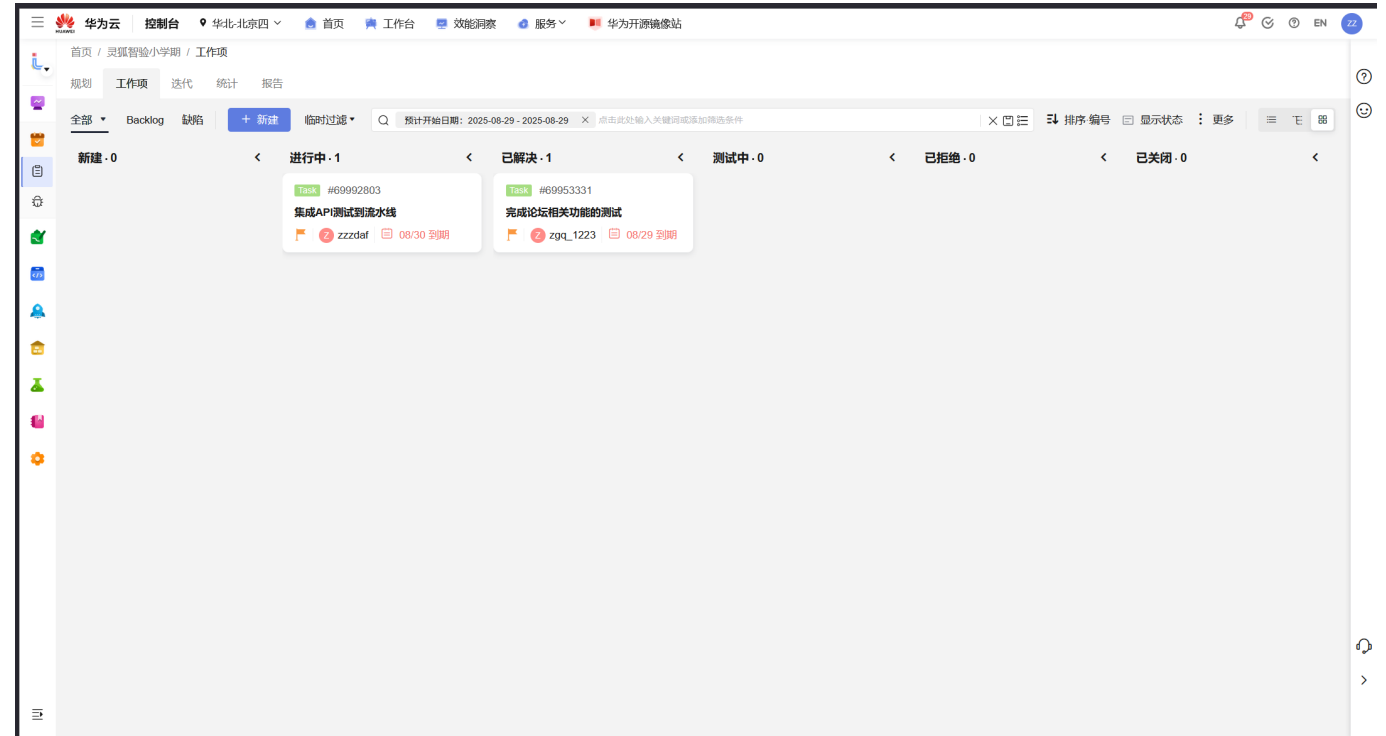
8.27:



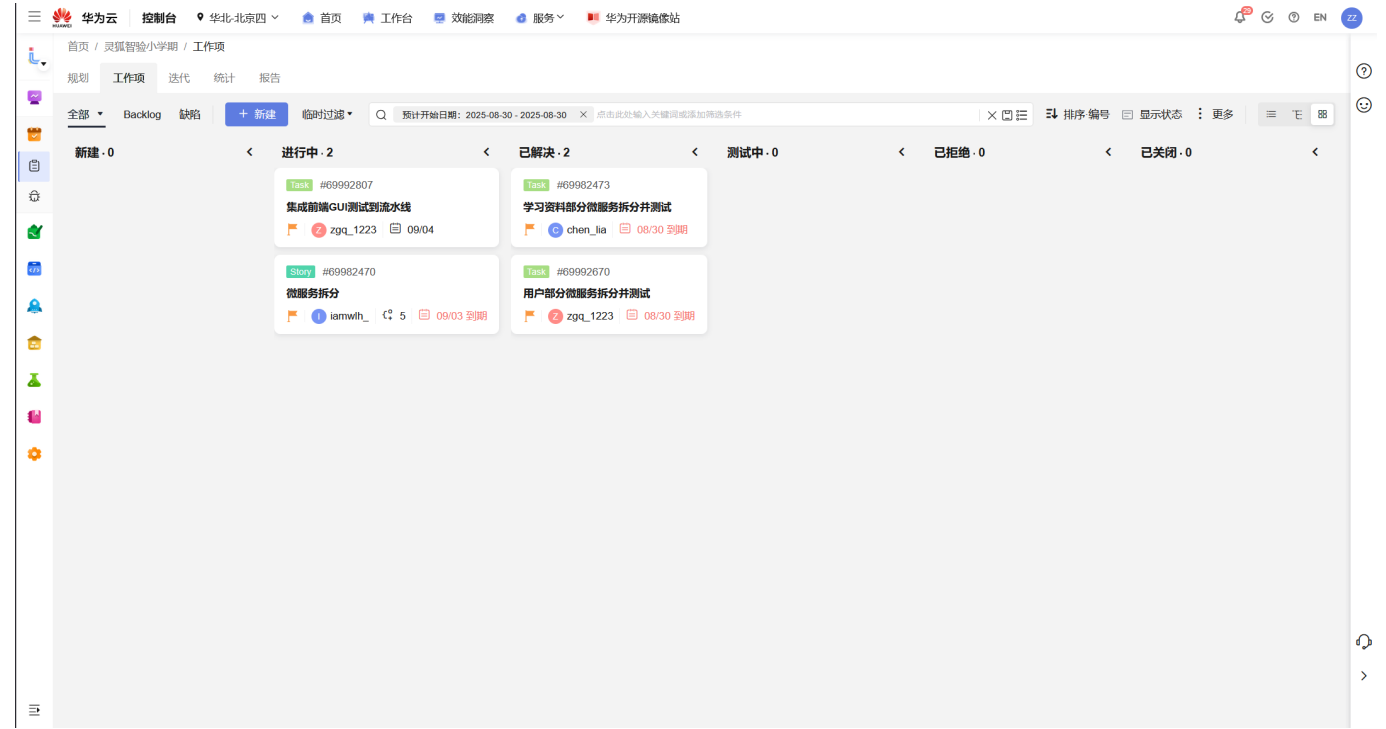
8.28:



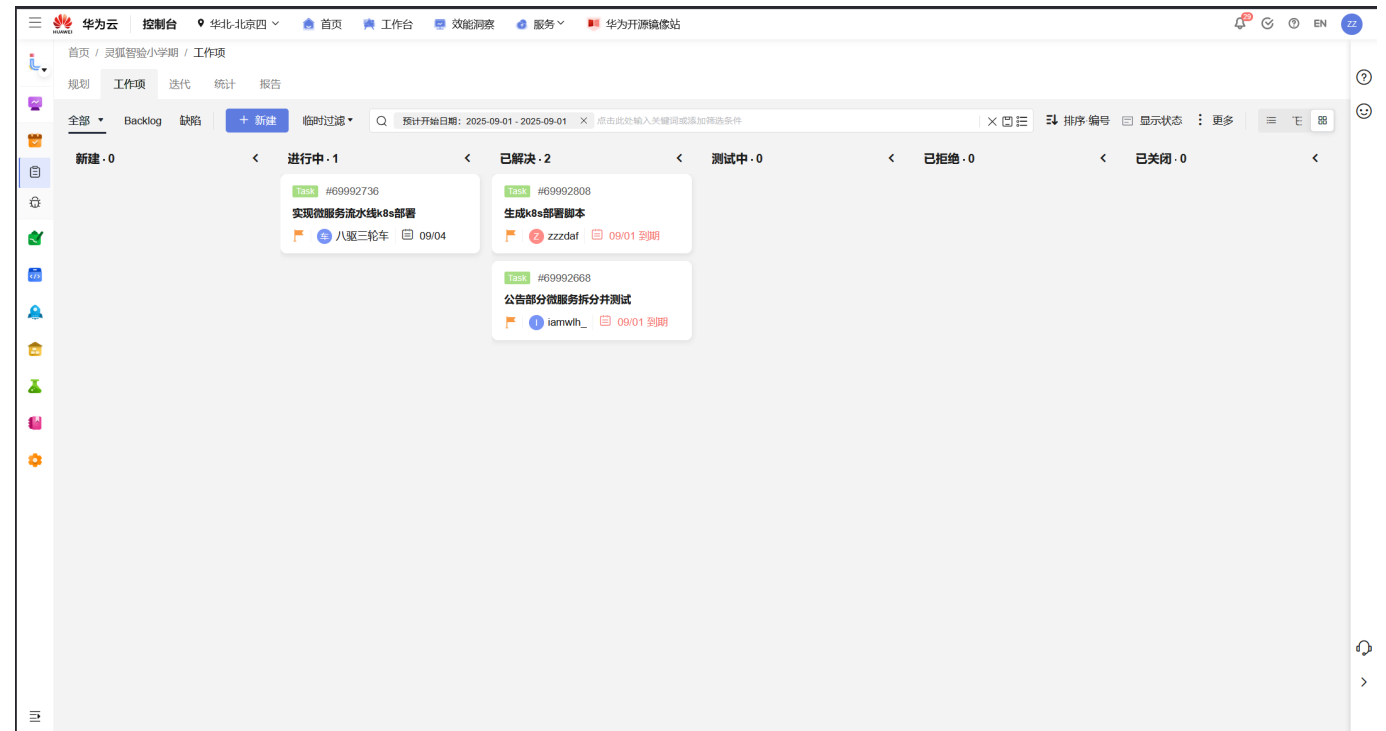
8.29:



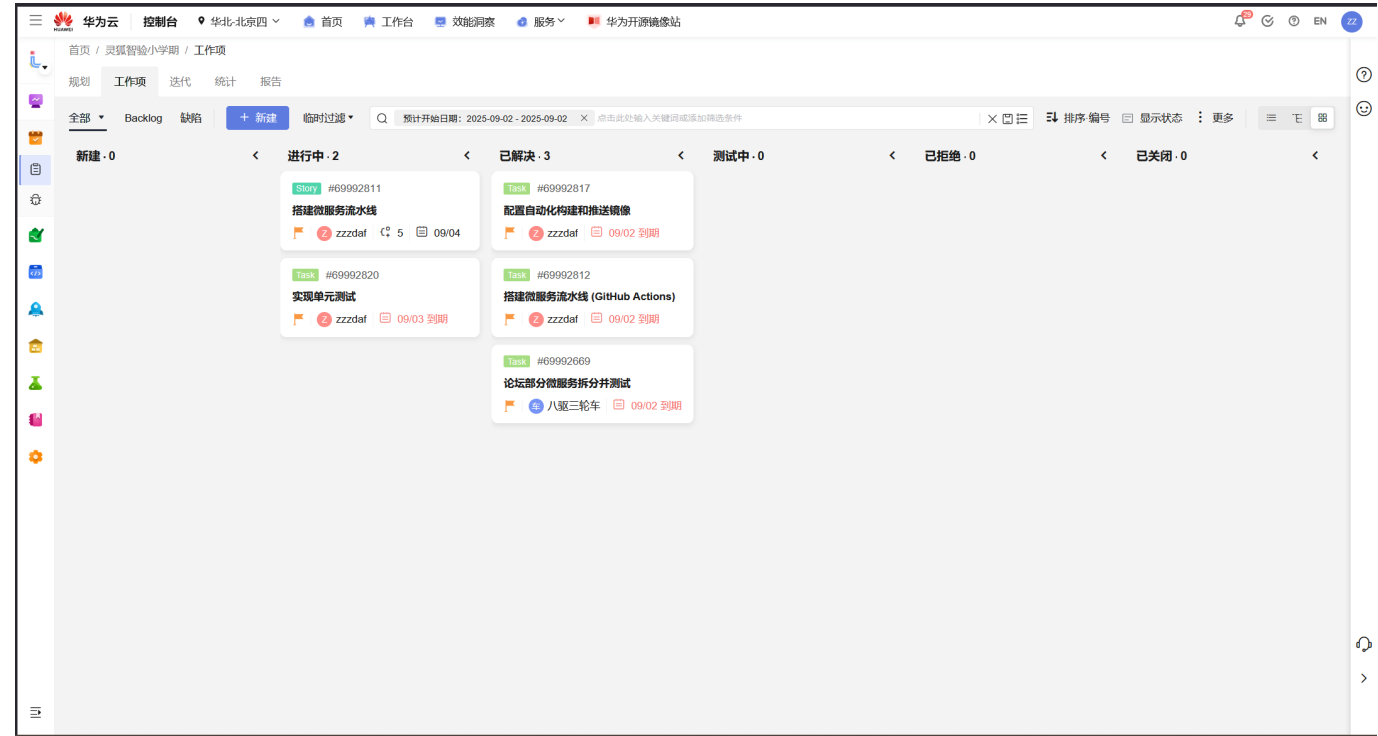
8.30:



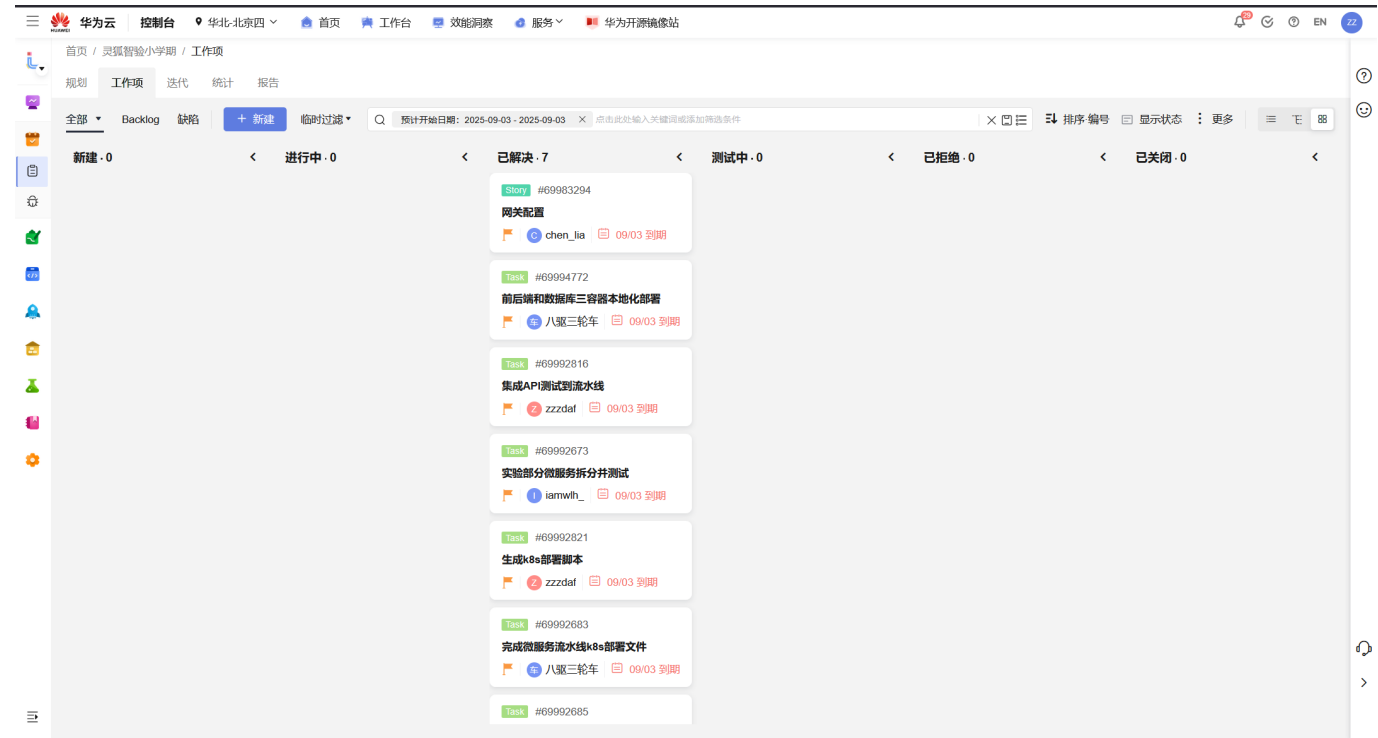
9.1:



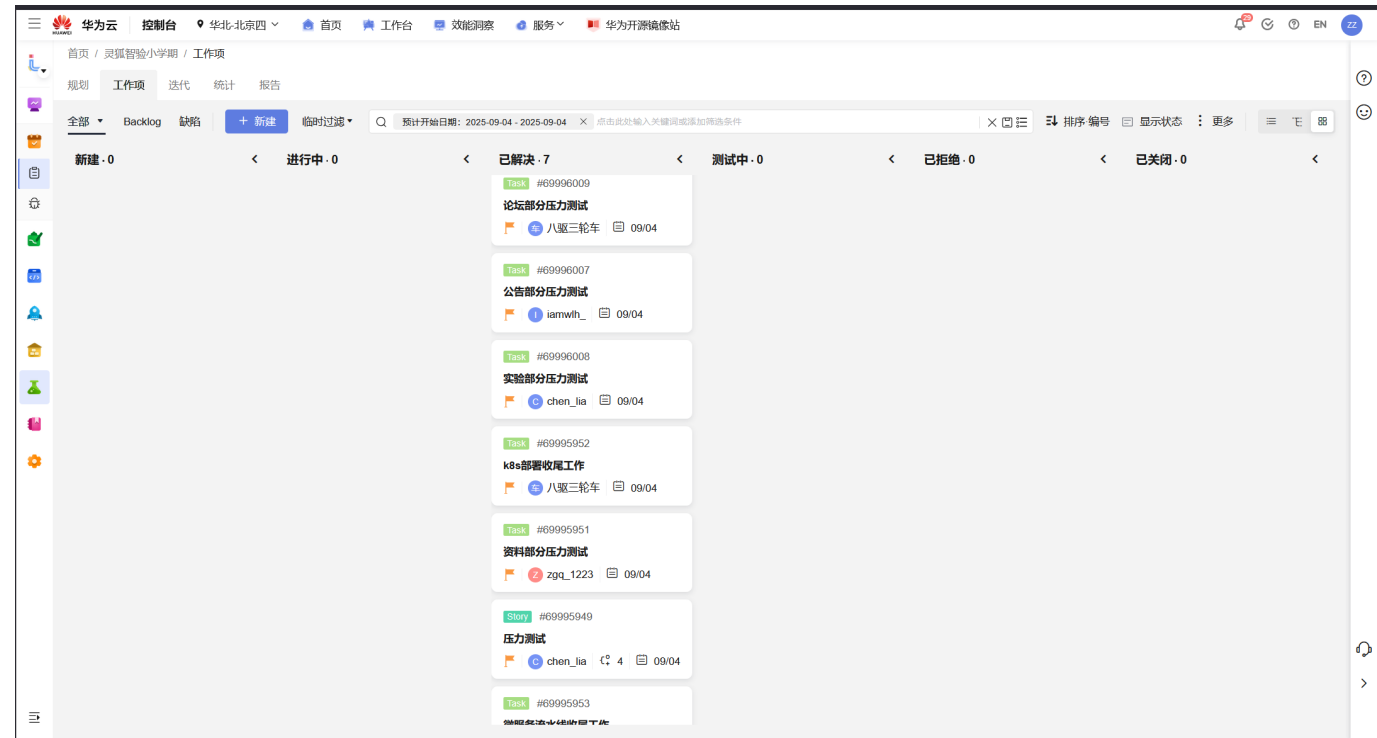
9.2:



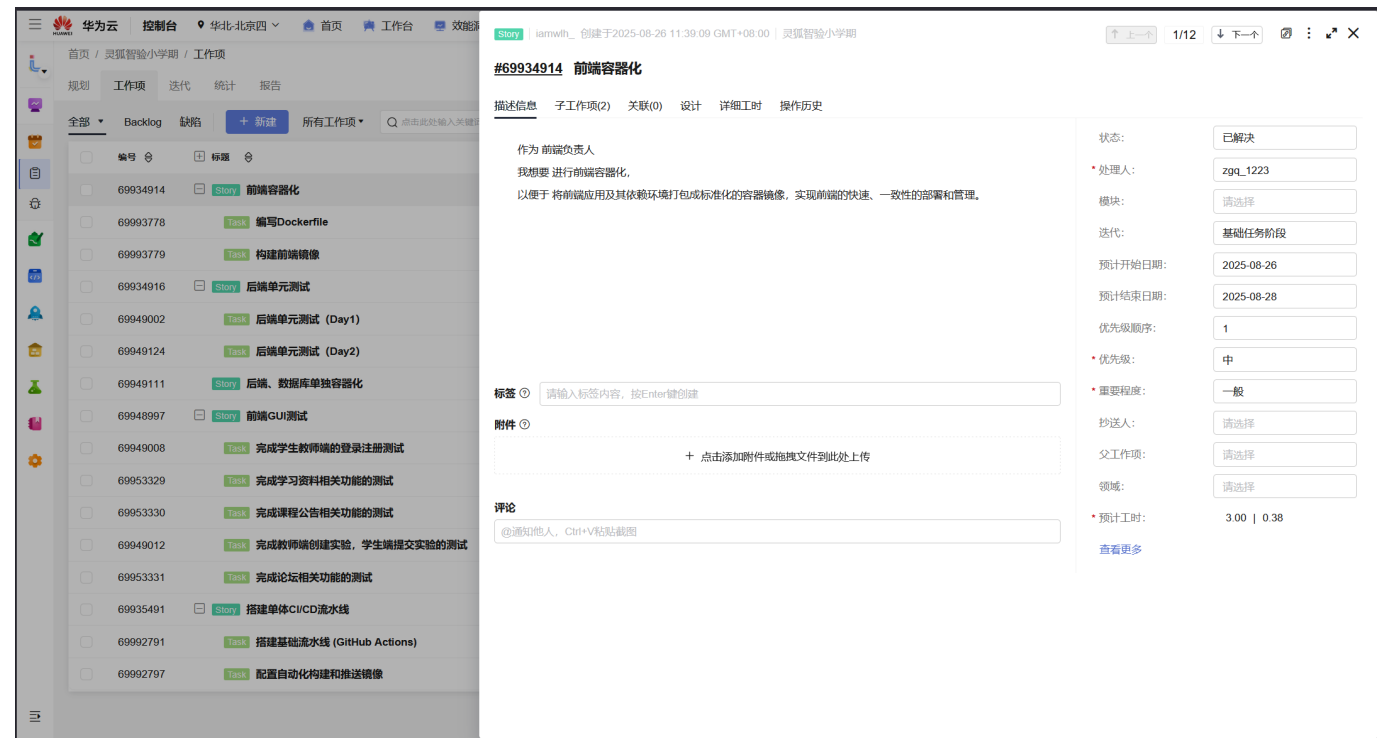
9.3:



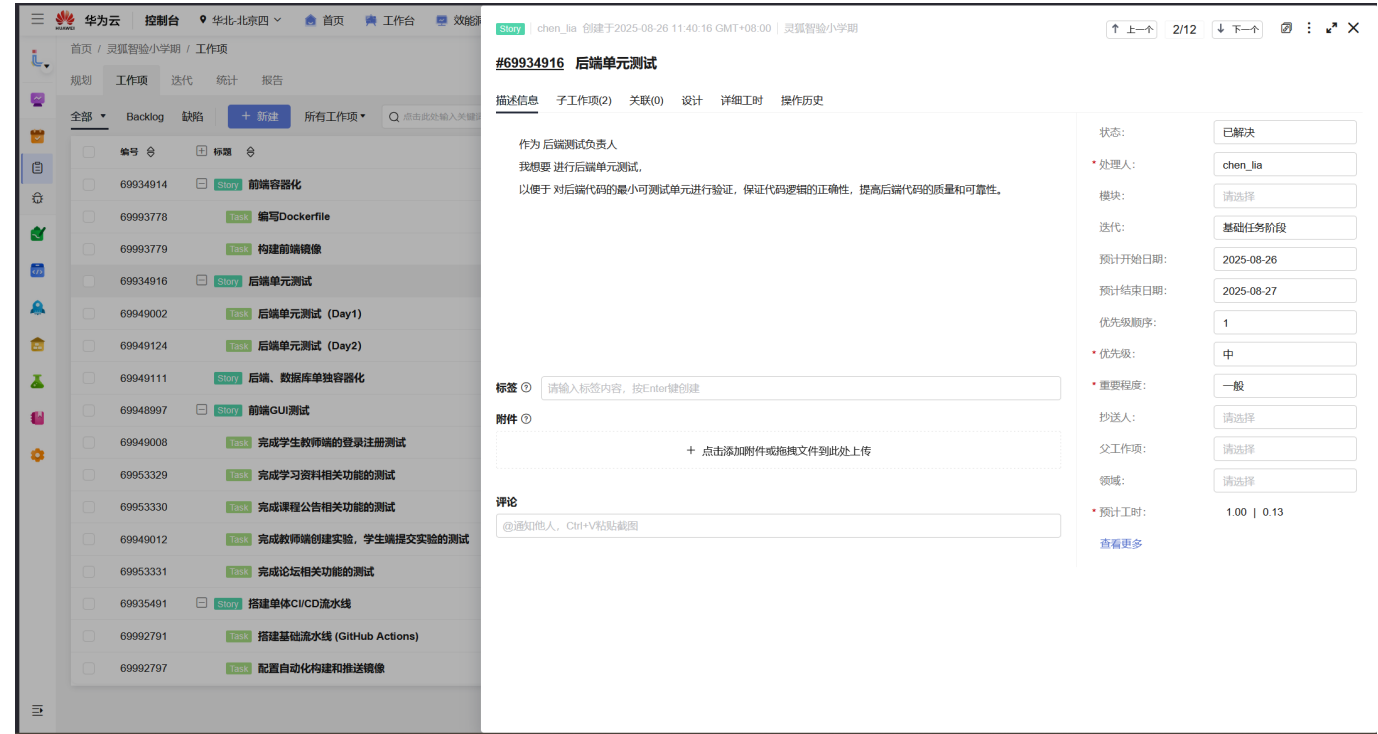
9.4:



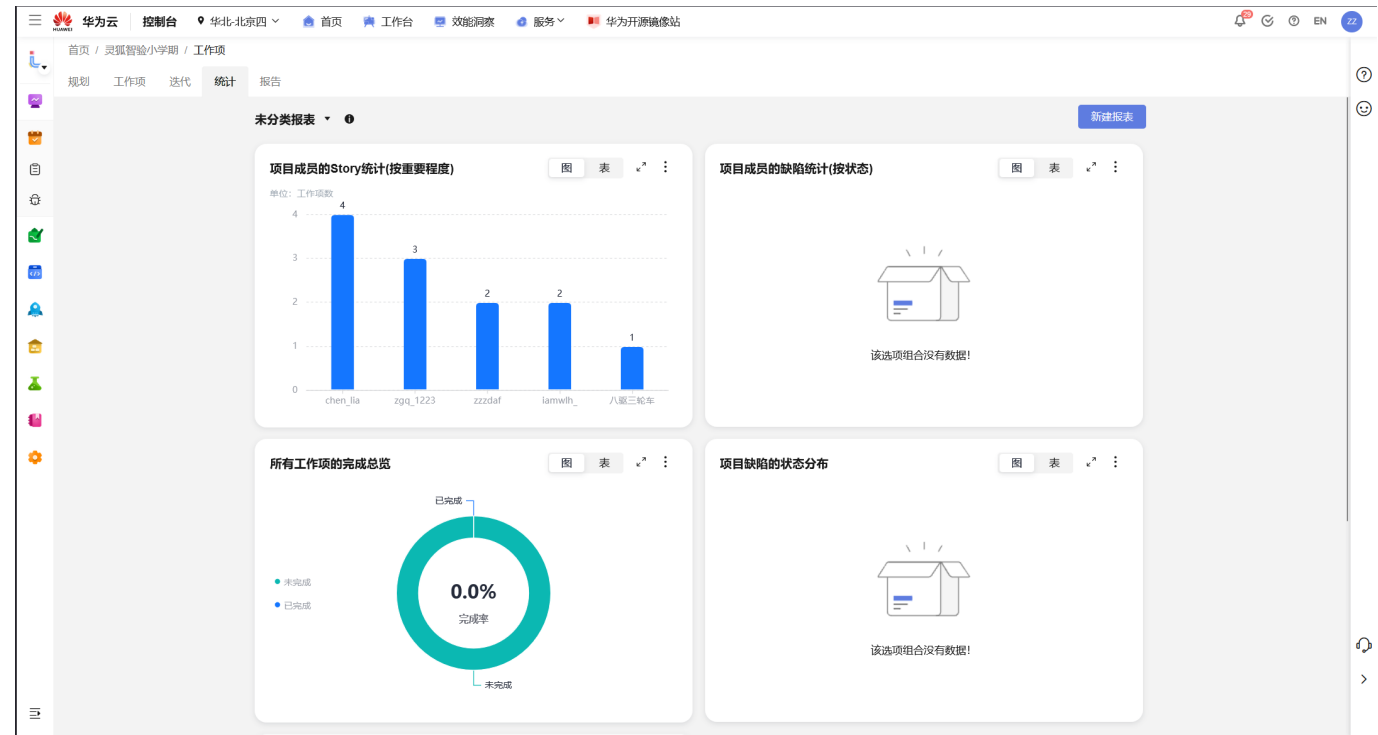
前端容器化：



后端单元测试：



项目统计：



任务使用的技术栈

前端单元测试：Jest + Vue Test Utils

后端单元测试：pytest

前端容器化：Node.js、Nginx、Docker、Kubernetes (K8S on Docker Desktop)

后端容器化：docker,docker-compose

流水线构建：Github Actions

本地容器: docker, docker-compose

前端GUI测试: Cypress

后端集成测试: Postman

压力测试工具: Locust

网关: nginx

代码异步测评: Celery, redis (消息中间件)

k8s本地化部署: Docker deployment service secret PVC

微服务本地化部署: nginx gateway

自动扩缩容: HPA

压力测试: Locust

微服务划分策略和理由

1. 数据库模型调整:

- 移除跨服务依赖的ForeignKey字段
- 改为存储原始ID字段(author_id, experiment_id)
- 使用JSONField存储读者列表(readers)

2. 认证处理:

- 使用SIMPLE_JWT进行身份验证
- 从JWT token解析用户信息而非直接查询用户表
- 基于用户角色(TEACHER/STUDENT)实现权限控制

3. 跨服务调用:

- 在序列化器中预留位置调用用户服务API获取用户信息
- 在序列化器中预留位置调用实验服务API获取实验信息

4. 独立数据库:

- 使用专用的微服务数据库
- 移除对User和Experiment模型的直接依赖

5. JSONField操作:

- 使用__contains进行JSON字段查询
- 使用列表操作代替多对多关系

6. 微服务特有配置:

- 独立SECRET_KEY
- 专用API路由(/api/notice/)
- 独立数据库迁移系统

流水线结果截图

单体系统流水线:



微服务化后流水线:

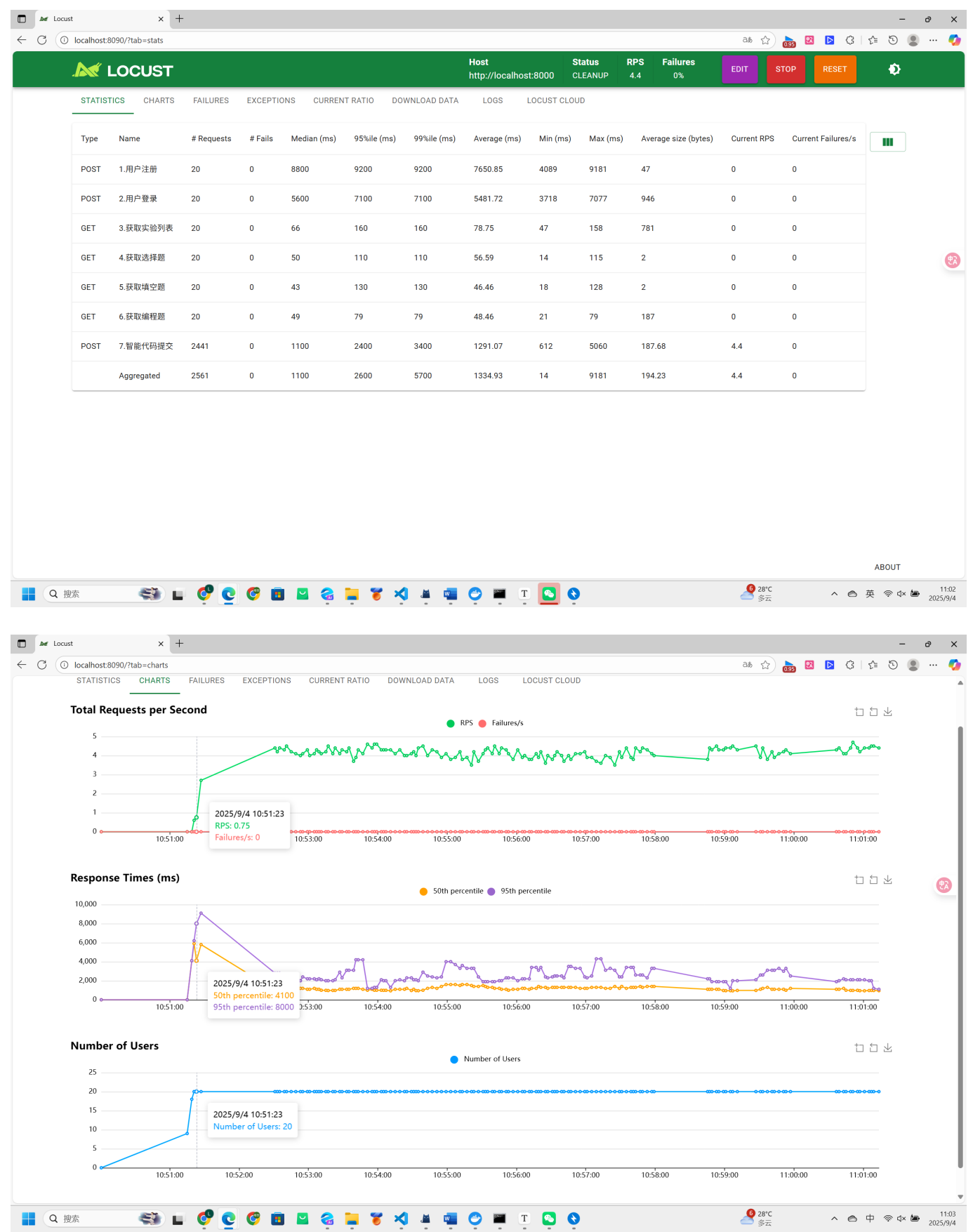


扩缩容、降级服务等功能的样例和记录

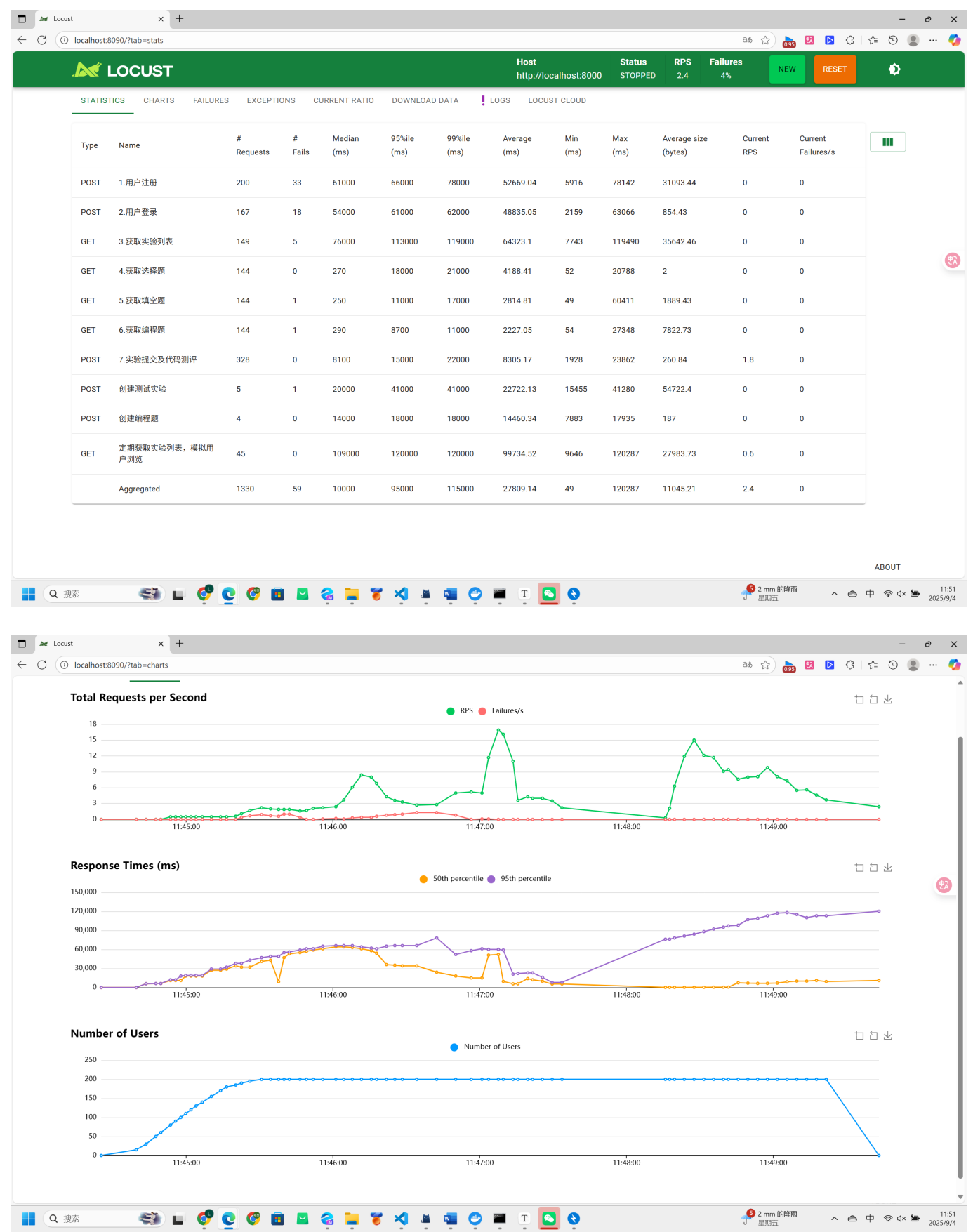
```
PS D:\Code\HTML\大二暑假小学期\group-platform\k8s> kubectl get hpa your-microservice-hpa --watch
NAME                                REFERENCE                                TARGETS      MINPODS  MAXPODS  REPLICAS  AGE
your-microservice-hpa              Deployment/user-service                  cpu: 20%/60% 2          10        2         3m58s
your-microservice-hpa              Deployment/user-service                  cpu: 22%/60% 2          10        2          4m
your-microservice-hpa              Deployment/user-service                  cpu: 20%/60% 2          10        2          5m
your-microservice-hpa              Deployment/user-service                  cpu: 21%/60% 2          10        2          6m
your-microservice-hpa              Deployment/user-service                  cpu: 84%/60% 2          10        2          7m
your-microservice-hpa              Deployment/user-service                  cpu: 120%/60% 2          10        3          8m
your-microservice-hpa              Deployment/user-service                  cpu: 19%/60% 2          10        4          9m
your-microservice-hpa              Deployment/user-service                  cpu: 20%/60% 2          10        4         11m
your-microservice-hpa              Deployment/user-service                  cpu: 21%/60% 2          10        4         12m
your-microservice-hpa              Deployment/user-service                  cpu: 21%/60% 2          10        4         13m
your-microservice-hpa              Deployment/user-service                  cpu: 22%/60% 2          10        2         14m
your-microservice-hpa              Deployment/user-service                  cpu: 21%/60% 2          10        2         16m
your-microservice-hpa              Deployment/user-service                  cpu: 22%/60% 2          10        2         18m
```

性能差距截图和简单说明

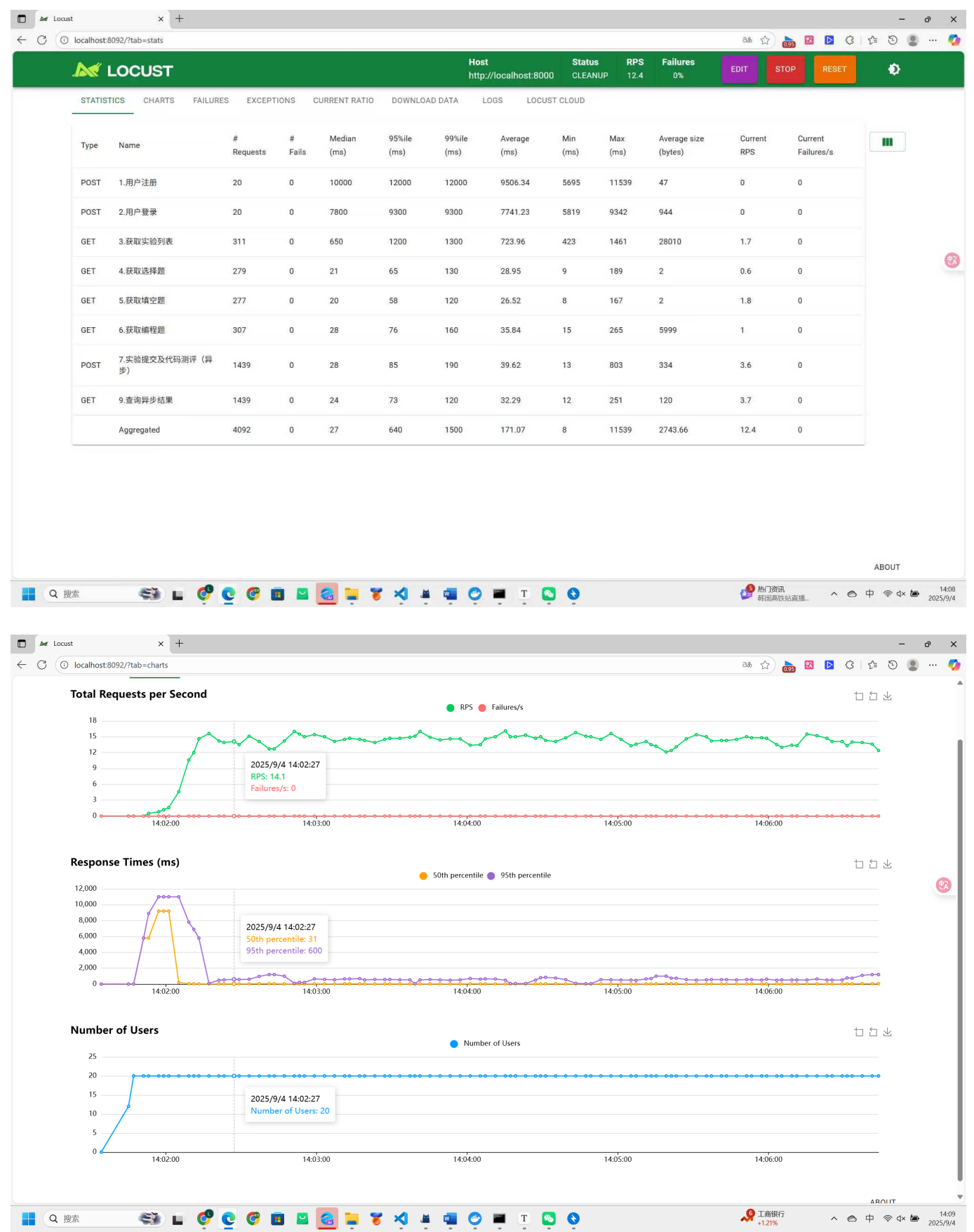
单体同步代码测评（用户数20；每秒启动4个新用户；运行时长10m）结果如下：



单体同步代码测评（用户数200；每秒启动5个新用户；运行时长5m）结果如下：



微服务异步代码测评（用户数20；每秒启动4个新用户；运行时长5m）结果如下：



使用Locust进行压力测试的结果：

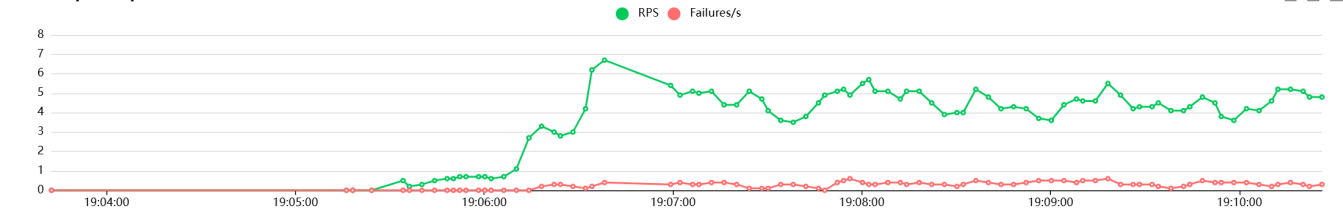
单体系统同步测评：平均响应时间为1300ms

微服务化且异步测评：平均响应时间为40ms

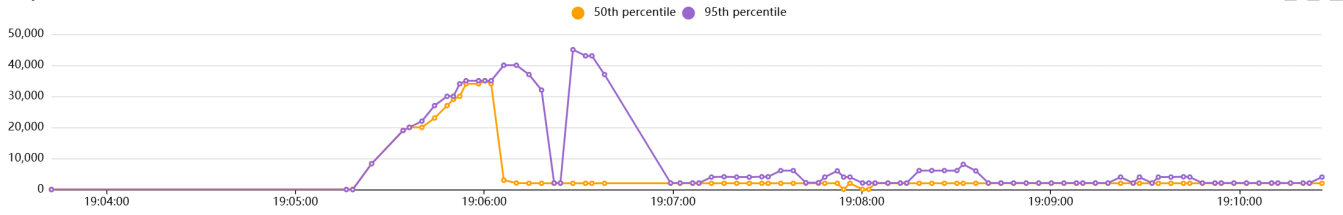
异步测评后每秒请求数是同步测评的4-5倍，异步测评响应时间更快更稳定。

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
POST	1.用户注册	20	0	23000	35000	35000	22677.76	8086	34763	47	0	0
POST	2.用户登录	20	0	38000	45000	45000	38105.73	30790	44981	877	0	0
POST	3.发帖(初始化)	20	0	2024.82	3000	3000	2144.44	2025	3032	344.75	0	0
POST	4.发帖	352	0	2100	2100	2100	2064.24	2020	2295	345.76	1.3	0
GET	5.获取帖子详情	166	0	2015.25	8100	10000	3219.52	2015	10111	461.7	0.7	0
POST	6.评论帖子	186	0	2018.59	2100	2100	2050.45	2019	2128	202.42	0.9	0
PATCH	7.点赞/取消点赞	179	0	22	43	64	24.09	11	65	30.24	1	0
PATCH	8.置顶/取消置顶	94	0	22	40	56	22.91	11	56	18.24	0.3	0
DELETE	9.删除帖子	162	87	18	44	62	21.44	8	71	22.56	0.6	0.3
	Aggregated	1199	87	2000	6100	37000	2427.8	8	44981	226.99	4.8	0.3

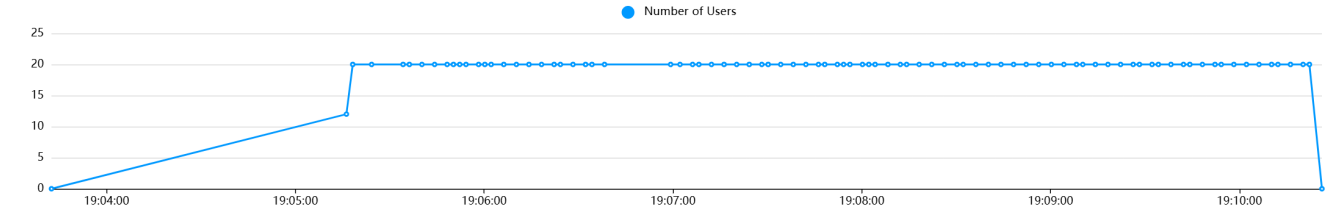
Total Requests per Second



Response Times (ms)

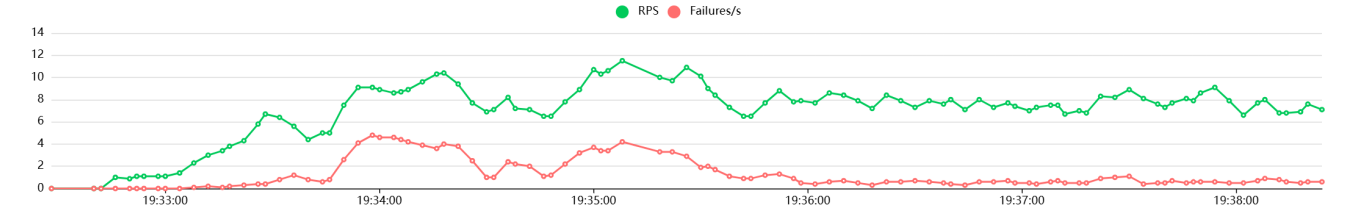


Number of Users

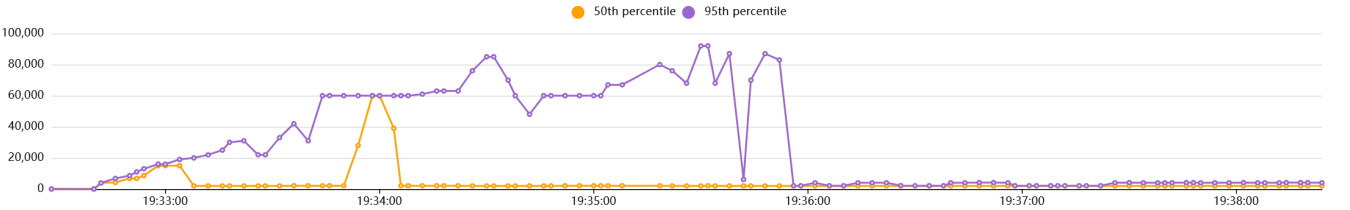


Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
POST	1.用户注册	200	142	60000	69000	85000	49428.67	3945	92139	131.45	0	0
POST	2.用户登录	200	167	60000	92000	92000	57647.66	14393	92360	272.81	0	0
POST	3.发帖(初始化)	33	0	2016.61	2100	2100	2041.4	2017	2143	346	0	0
POST	4.发帖	634	0	2100	2100	2200	2074.07	2016	2276	346.21	1.9	0
GET	5.获取帖子详情	316	0	2011.78	6100	8100	2976.42	2012	14120	437.41	1.5	0
POST	6.评论帖子	346	0	2014.92	2100	2100	2056.96	2015	2170	203	1.1	0
PATCH	7.点赞/取消点赞	314	0	28	64	120	34.31	9	263	30.24	1.1	0
PATCH	8.置顶/取消置顶	142	0	27	61	82	33.76	11	232	18.25	0.3	0
DELETE	9.删除帖子	323	153	24	64	98	29.93	4	401	19.89	1.2	0.6
Aggregated		2508	462	2000	60000	81000	9758.8	4	92360	214.81	7.1	0.6

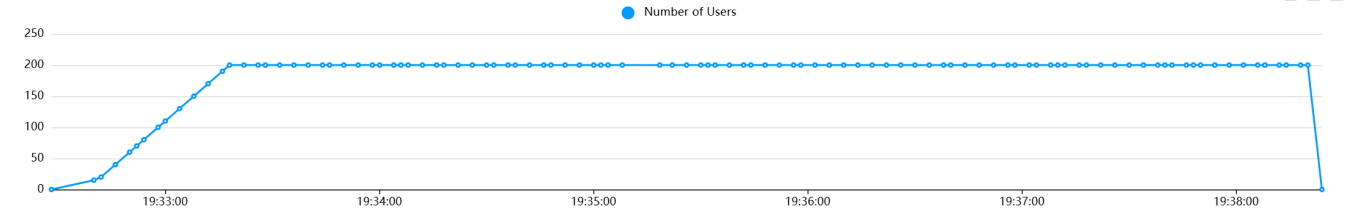
Total Requests per Second



Response Times (ms)



Number of Users



总结与评价

微服务拆分带来了革命性的性能提升：

指标	拆分前 (单体应用)	拆分后 (微服务)	变化评价
95th 响应时间	~7000 ms	极低 (例如 ~200ms)	✔ 巨大提升 (延迟降低35倍以上)
50th 响应时间	~2000 ms	极低 (例如 ~50ms)	✔ 巨大提升 (延迟降低40倍)
吞吐量 (RPS)	低, 受限于用户数	高, 且更稳定	✔ 显著改善
错误率	高	极低	✔ 显著改善
稳定性	极不稳定, 波动大	非常稳定	✔ 根本性改善
可扩展性	极差	良好	✔ 根本性改善

权重分配

成员	权重
王露函	1
陈利雅	1
张管倩	1
张宸菲	1
马浩龄	1